

Part 2 of 2

Network Protocols and Smart Sensors: An Introduction to Wireless Sensor Networks

Wireless sensor networks (WSNs) have seen explosive growth in recent years because of their independent sensing capabilities. The first part of this article gave basics of the technologies used in WSNs. This concluding part covers network protocols and types of sensors generally used

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A wireless sensor network is a communication system which senses and gathers information from a certain area and sends it to where it is required. In such networks the communication system requires networking protocols which are efficient, reliable, scalable and secure.

Fundamental differences in sensor networks which affect protocols are:

Data sink(s). That is, the nature of data sink(s). For example, whether the end user is embedded in sensor network or access points

Sensor mobility. This aspect may influence protocols at the networking level as well as at localisation service

Sensor resources. Such as computing ability

Traffic pattern. For example, whether data is generated continuously during environmental monitoring

The factors to be considered while designing WSN protocols are:

1. Routing decisions to be undertaken considering the importance of energy resources in the network.

2. Communication channels often exist between events and sinks. Since sink nodes are typically used for overall description of the environment rather than explicit readings from the individual sensor devices, the communication in sensor networks is normally referred to as data-centric rather than address-centric, and the data may be aggregated locally rather than collecting all the raw data sent to the sink(s).

3. Sensors have the knowledge of

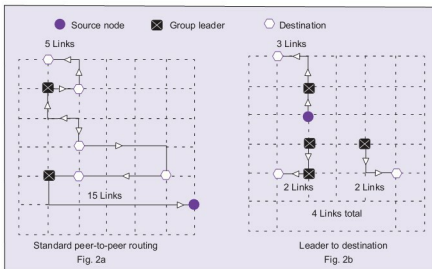


Fig. 2: Multicast routing reduces message path length and improves efficiency

their own location to usefully assess their data. The location information can be utilised for routing purposes.

4. If a sensor network is well connected, the topology control service should be used in conjunction with the normal routing protocols.

Multiple access protocols

When multiple nodes desire to transmit, protocols are needed to avoid collisions and loss of data. In frequency division multiple access (FDMA), different nodes have different carrier frequencies. FDMA also requires additional hardware and intelligence at each node. In code division multiple access (CDMA), a unique code is used by each node to encode its messages. However, this increases complexity of the transmitter and receiver.

In time division multiple access

(TDMA), the RF link is divided on time axis with each node given a predetermined time slot it can use for communication. This decreases the sweep rate, but a major advantage is that it can be implemented in software. All nodes require accurate, synchronised clocks for TDMA.

Medium access control protocols

Medium access control (MAC) protocols focus on reducing the idle power consumption by setting the sensing transmitters to sleeping mode as often as possible. MAC protocols have been designed for ad-hoc networks which primarily focus on optimising fairness and throughput efficiency with less emphasis on energy conservation. Some protocols like IEEE 802.11 eliminate the waste caused by